

Nathaniel Bottman

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Machine learning researcher working on evaluation and reinforcement-learning post-training for multi-step reasoning in large language models (LLMs). PI of a DARPA-funded research program (“An Operad-Based Framework for Deep Learning,” \$836K). Introduced *operadic consistency*, a label-free signal that checks a model’s answer against its own decomposition of the problem. Previously a research group leader at the Max Planck Institute for Mathematics.

EXPERIENCE

Incubilate – CEO & Principal Scientist

Seattle, WA, July 2025–present

- PI of “An Operad-Based Framework for Deep Learning” (\$836K, DARPA AIQ program): operads as the mathematical structure of question decomposition, and label-free grading of multi-step reasoning.
- Developed *operadic consistency* (OC): tracks accuracy at $r = 0.86$ – 0.94 across twelve LLMs on four multi-hop QA benchmarks, predicts a held-out model’s accuracy to 3.0 pp with no labels, and improves selective prediction over a tuned self-consistency baseline at equal cost.
- Extended OC to frontier reasoning models (DeepSeek-R1, Kimi-K2.6, GLM-5.1) by extracting the decomposition from the model’s own chain of thought and re-deriving it; positive lift on all 16 selective-prediction cells tested.
- Integrated OC as an accuracy predictor in a DARPA evaluation harness, incl. a trace-extracted OC node for reasoning models; stack spans open-weights (4B–671B) and API models.
- In progress: GRPO post-training for honest refutation on an exactly decidable reasoning task.

Max Planck Institute for Mathematics – Research Group Leader (W2)

Bonn, Jan 2021–Jun 2025

- Led a research group in symplectic geometry; supervised six postdoctoral researchers and multiple student theses; served on the institute’s Scientific Committee (hiring).

Postdoc positions: U. Southern California (2019–20), Inst. for Advanced Study (2016–19), Northeastern (2015–16).

SELECTED PUBLICATIONS

1. **Operadic consistency: a label-free signal for compositional reasoning failures in LLMs.** N. Bottman, Y. Liu, K. Richardson. arXiv:2606.13649. NeurIPS 2026 (poster).
Code: github.com/natebottman/operadic-consistency-paper.
2. **Operads for compositional reasoning in LLMs.** N. Bottman, K. Richardson. arXiv:2606.13634. Presented at the Compositional Learning and the Combining Theory and Benchmarks workshops, ICML 2026.

Twenty further publications in symplectic geometry, combinatorics, and applied mathematics, in venues including the *Bulletin of the AMS*, *Selecta Mathematica*, *Algebraic & Geometric Topology*, and *J. Symplectic Geometry*. These include a polytopal realization of higher-categorical associahedra found with OpenAI’s Astra model and formalized in Lean 4 (arXiv:2609.25316).

GRANTS & FELLOWSHIPS

- DARPA AIQ program, “An Operad-Based Framework for Deep Learning,” \$836K, 2025–present (PI).
- NSF Standard Grant DMS-1906220, \$144K, 2019–2022 (PI).
- NSF Mathematical Sciences Postdoctoral Research Fellowship, 2016–2019.
- NSF Graduate Research Fellowship, 2010–2015.

EDUCATION

Massachusetts Institute of Technology, Ph.D., Mathematics, 2015. Advisor: Katrin Wehrheim.

University of Washington, B.S. Mathematics and B.A. Slavic Languages & Literatures, 2010. Dean’s Medal (top graduating senior in the natural sciences).

SERVICE & COMMUNITY

Co-organizer, Combining Theory and Benchmarks Workshop, ICML 2026 (Seoul); co-organizer of research seminars and workshops since 2012, incl. the Western Hemisphere Virtual Symplectic Seminar (2020–23). 45+ invited talks.

TECHNICAL

Python (LLM evaluation pipelines; open-weights and API inference; GRPO reinforcement learning), PyTorch, L^AT_EX.